



UNIVERSITY OF COLOMBO, SRI LANKA



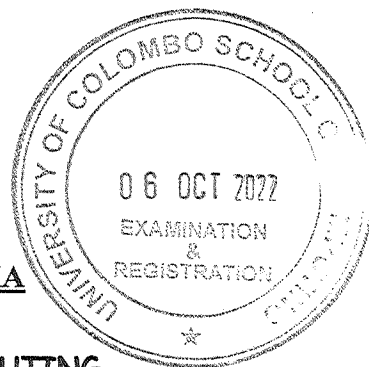
UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination – Semester I - 2022

SCS1205 – Computer Systems

TWO (2) HOURS



234

To be completed by the candidate

Index No:

Important Instructions to candidates:

1. The medium of instruction and question is **English**.
2. Note that questions appear on both sides of the paper.
3. If a page or part of this paper is not printed, please inform the supervisor immediately.
4. Write your index number on every page of the Question paper.
5. Answer **ALL** questions.
6. There are **04** questions in **16** pages.
7. All **04** questions carry **equal marks**.
8. Write your answers and index number **clearly and readable** manner.
9. In questions where justifications, reasons, and steps of calculations are specifically asked, the student must provide the same in the answer. **Failure to provide** the above as required in the question will result in that question or part being awarded **zero marks**.
10. Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are **not allowed**.
11. **Non-Programmable** calculators are **allowed**.

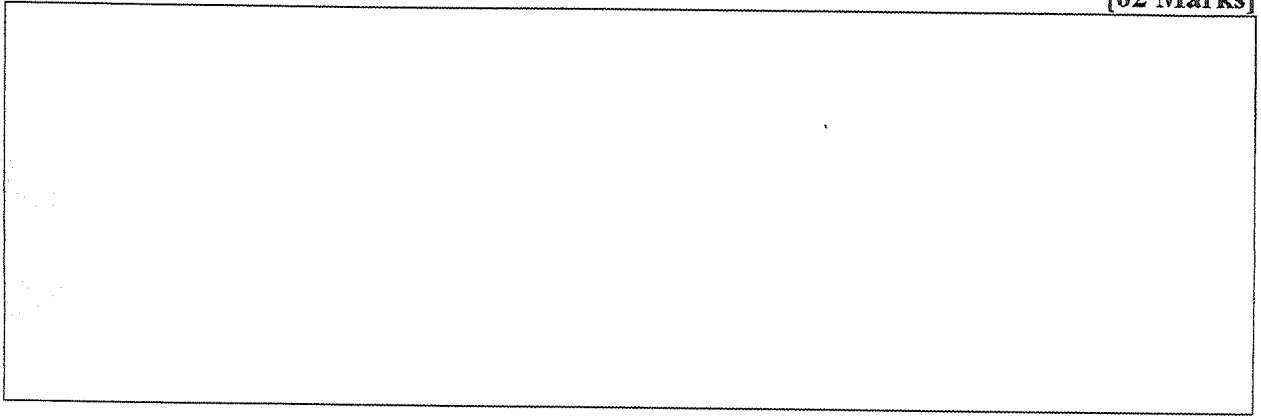
For Examiner's use only

Question No	Marks
1	
2	
3	
4	
Total	

Question 1

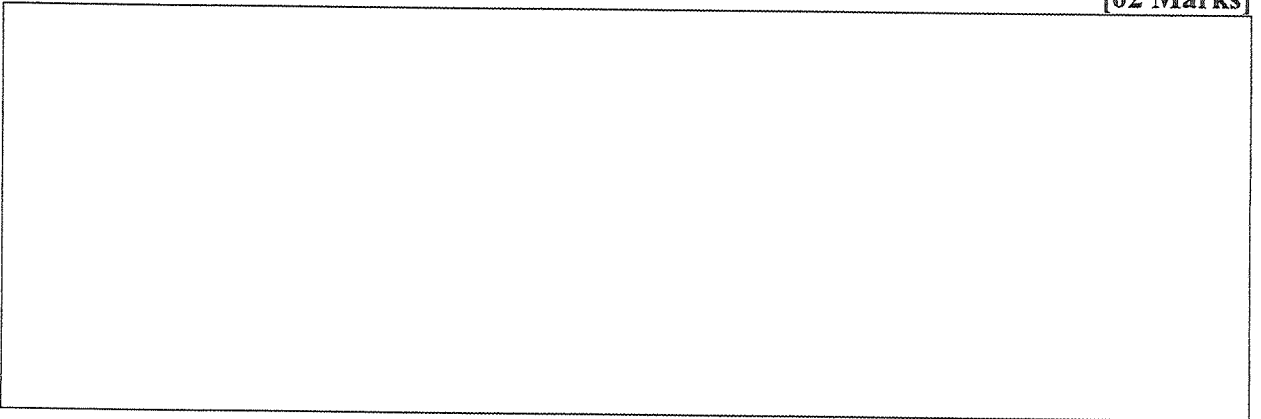
A. In the past, there were multiple generations of computers produced. What are meant by *zeroth generation* computers and provide a one (01) example for such computers.

[02 Marks]



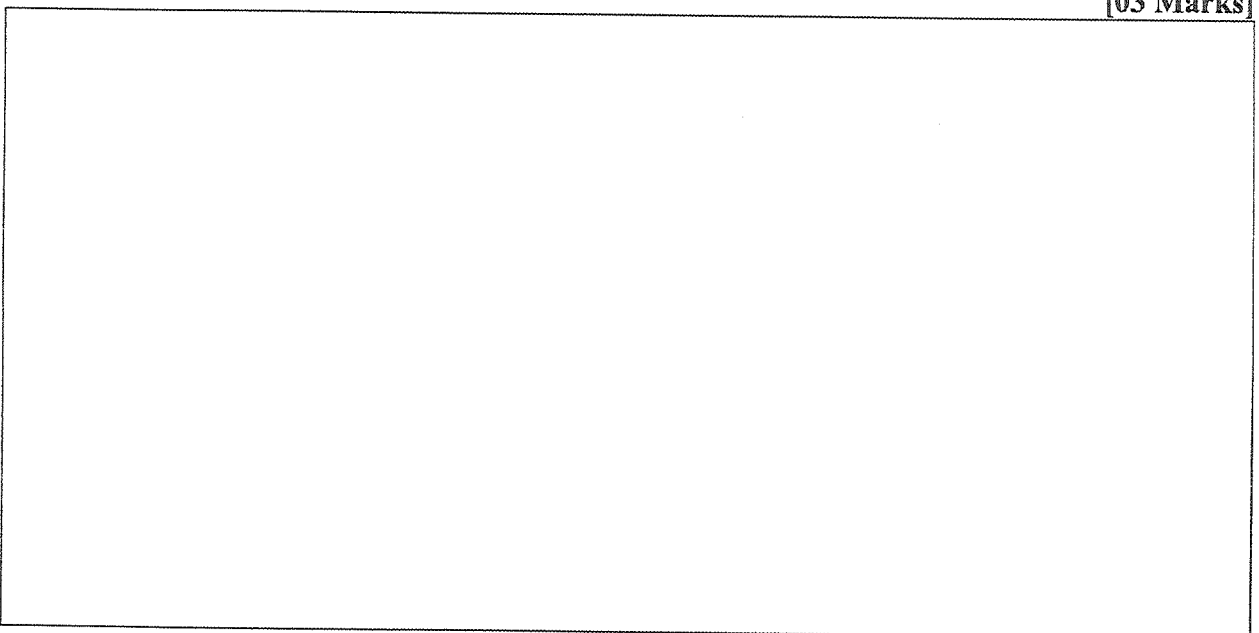
B. What will be the *sixth generation* of a computers and why do we need such computers?

[02 Marks]



C. What is meant by the *Moore's law* in computing? What is the fundamental rule behind the *Moore's law*?

[03 Marks]

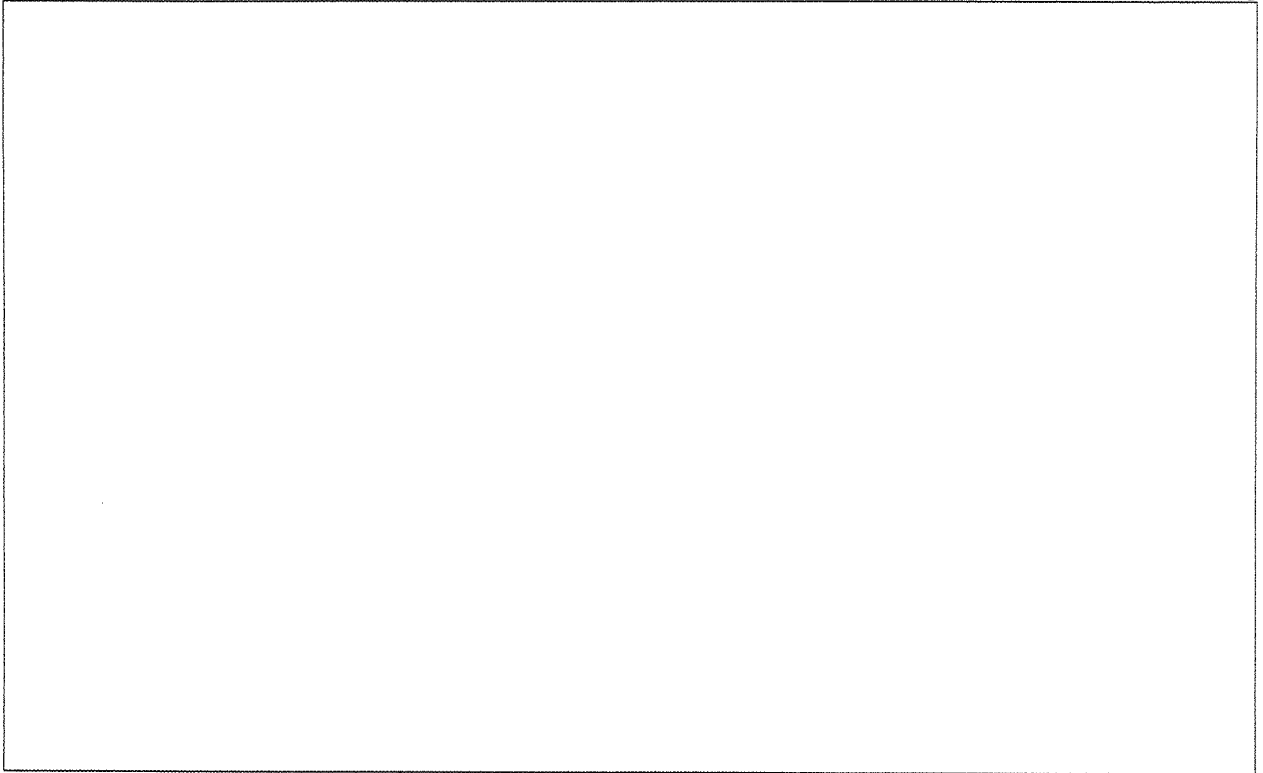


D. Assume that a sign bit, a 6-bit biased *exponent*, and a 9-bit *mantissa* are all part of a 16-bit floating point representation.

- i. What is the decimal equivalent of the 16-bit floating number below? Show your steps in finding the answer.

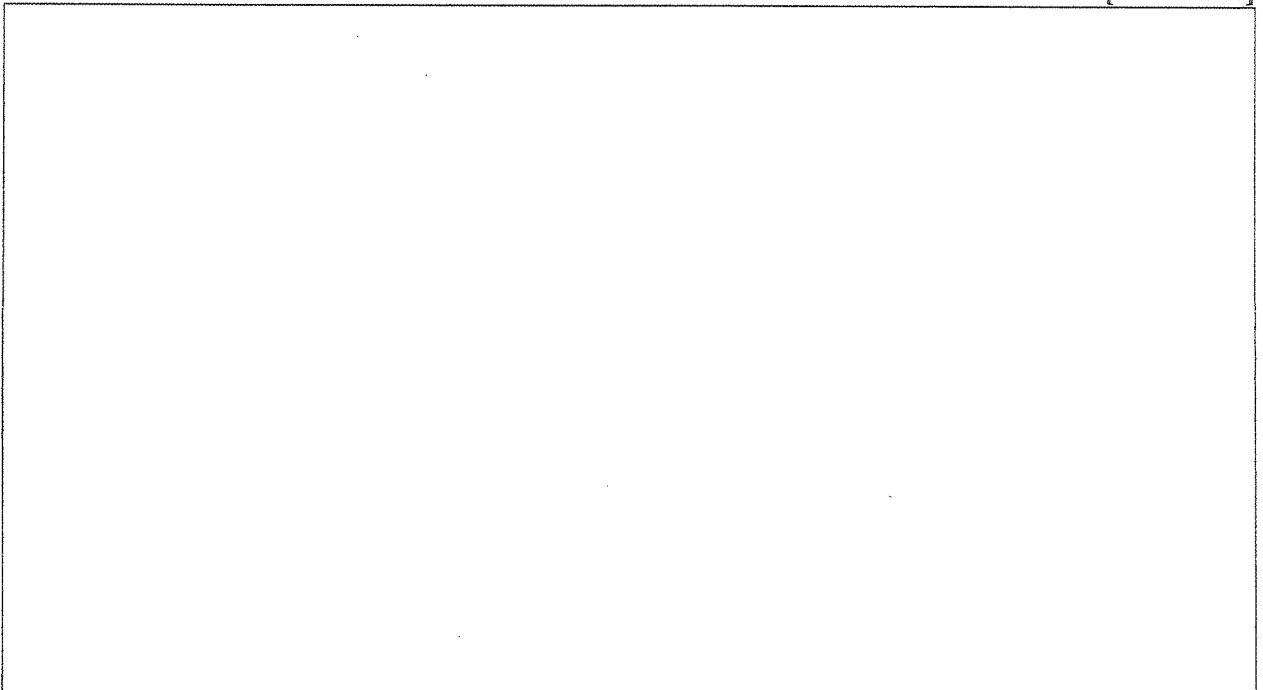
1 101101 111100000

[04 Marks]



- ii. What is the 16-bits floating point representation of the decimal value +52.125 ? Show your calculations and steps briefly.

[04 Marks]

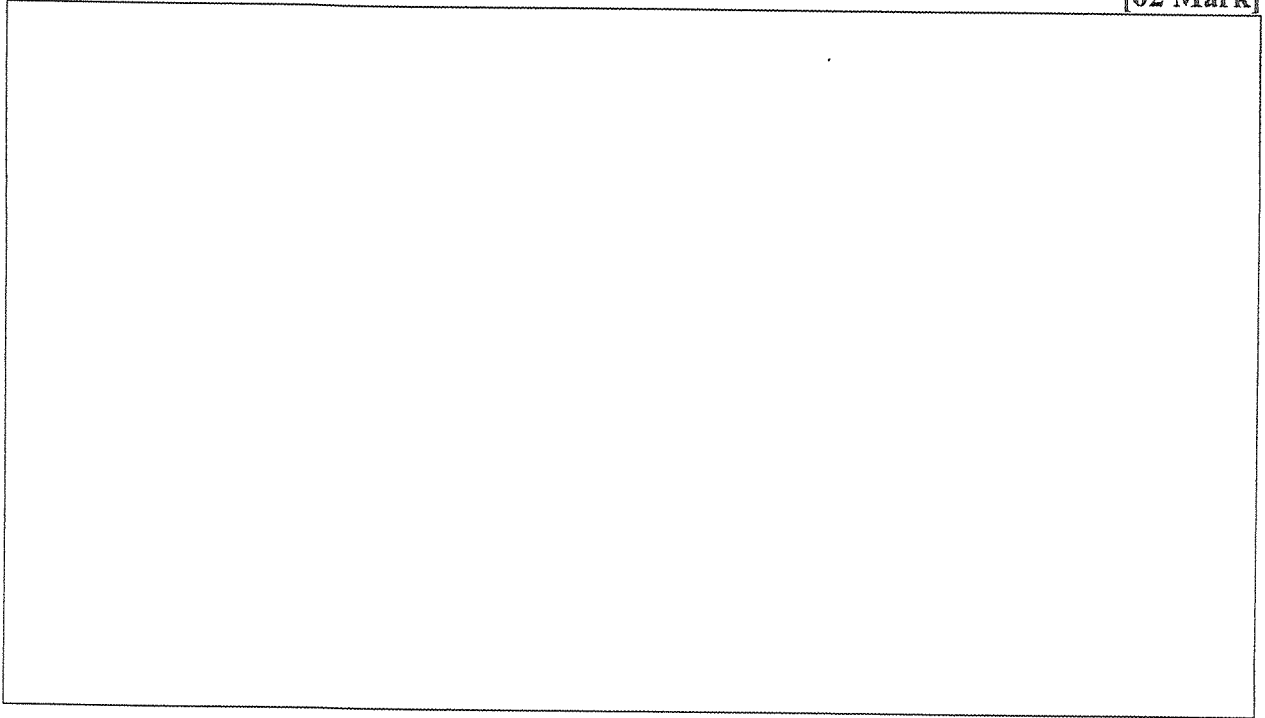


Index No:

E. Answer to the following questions assuming a floating-point representation with a *sign bit*, a 6-bit biased *exponent*, and a 9-bit *mantissa*.

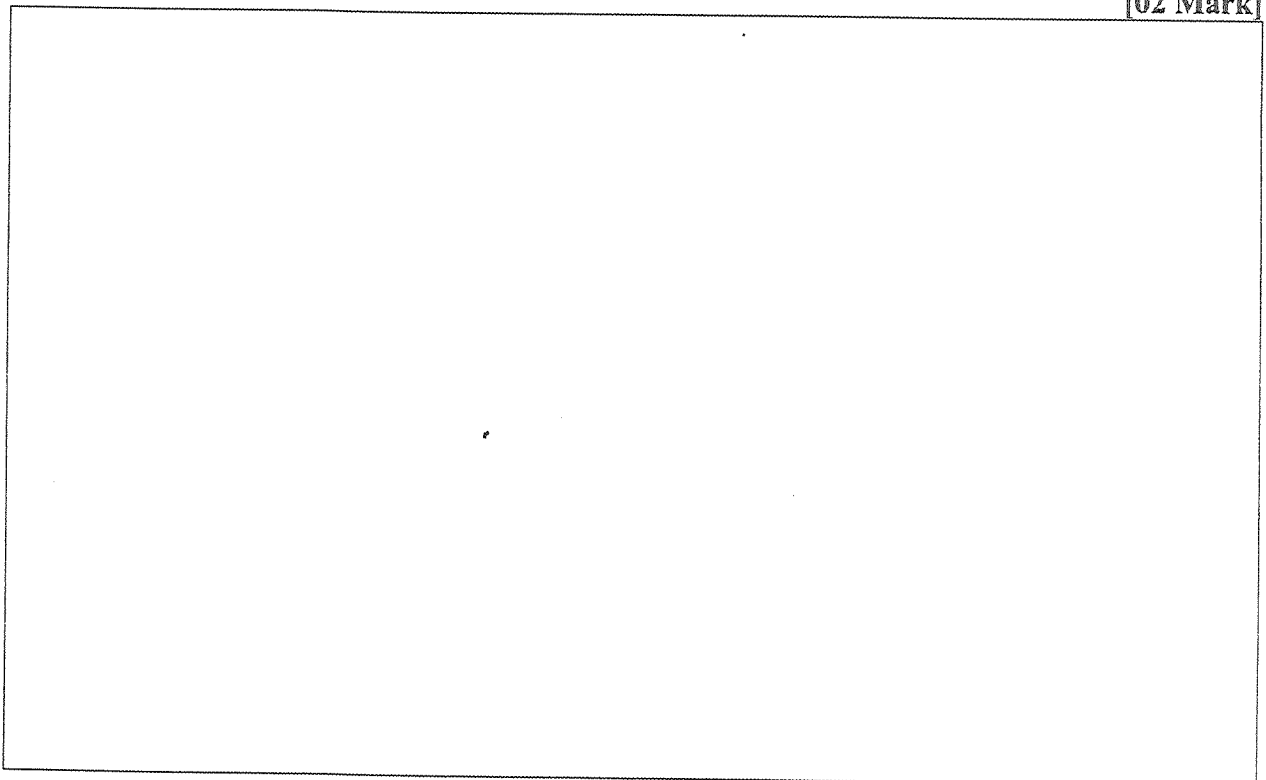
- i. What is the **largest positive number** that can be represented? Show your steps in finding the answer.

[02 Mark]



- ii. What is the **largest negative number** (i.e., negative number with largest magnitude) that can be represented? Show your steps in finding the answer.

[02 Mark]



- iii. What is the **smallest positive number** that can be represented? Briefly show your steps in finding the answer.

[02 Mark]

- iv. What is the **smallest negative number** (i.e., negative number with smallest magnitude) that can be represented? Briefly show your steps in finding the answer.

[02 Mark]

- v. What is meant by *positive underflow* and what the positive underflow range in this case?

[02 Marks]

Question 2

A. Consider the following Boolean algebraic expression to answer the questions below.

$$F = \bar{A}.\bar{B}.\bar{D} + \bar{A}.(\bar{B}.C.D + B.\bar{C}.\bar{D}) + (\bar{A}.B + A.\bar{B}).C.\bar{D}$$

- i. Convert the expression F into a form of *Sum of Minterms*.
Note the term “*Sum of Minterms*” in the question.

[02 Marks]

- ii. Complete the truth table below by filling the column F for the expression F given above.

[04 Marks]

Minterm Index	A	B	C	D	F
0	0	0	0	0	
1	0	0	0	1	
2	0	0	1	0	
3	0	0	1	1	
4	0	1	0	0	
5	0	1	0	1	
6	0	1	1	0	
7	0	1	1	1	
8	1	0	0	0	
9	1	0	0	1	
10	1	0	1	0	
11	1	0	1	1	
12	1	1	0	0	
13	1	1	0	1	
14	1	1	1	0	
15	1	1	1	1	

- iii. It has been confirmed that, whenever both A and C inputs are in Boolean logic state **ONE (1)**, B and D inputs are produced **only** in equivalent states (equivalent states means either B and C both are **ONE** or **ZERO**). Therefore, a *minterm* with both A and C are in Boolean logic state **ONE (1)**, B and D are in mutually opposite Boolean logic states will never happen. Accordingly, list all *minterm(s)* that are *Don't care conditions* for this description.

[04 Marks]

- iv. Considering answers for Question 2.i, 2.ii, and 2.iii, derive the simplified the expression in *Sum of Products form* for F using the K-Map given below. Indicate Boolean logic state **ONE** by 1, Boolean logic state **ZERO** by 0, and *Don't care conditions* by X. Clearly show your groupings on the K-Map.

[05 Marks]

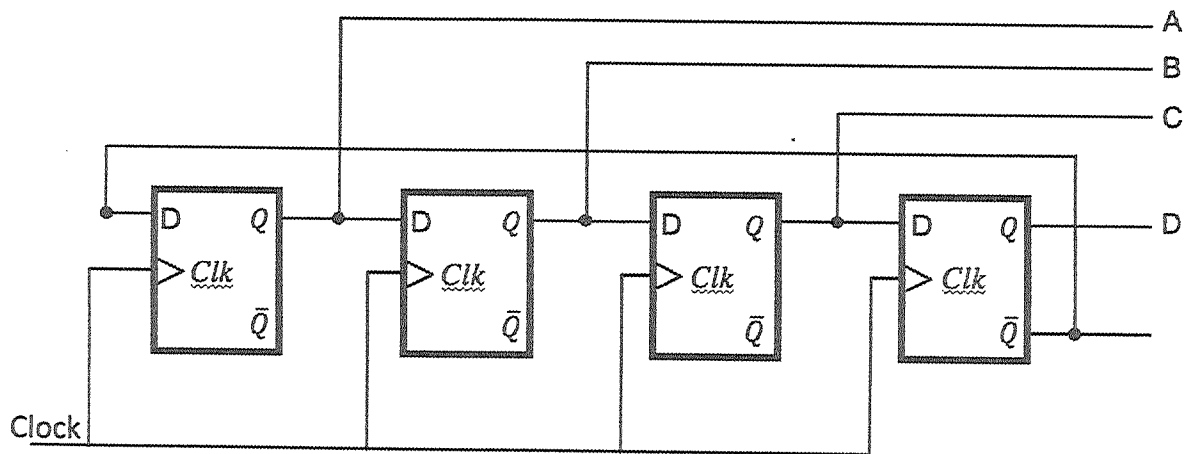
	$\bar{A}.\bar{B}$	$\bar{A}.B$	$A.B$	$A.\bar{B}$
$\bar{C}.\bar{D}$				
$\bar{C}.D$				
$C.D$				
$C.\bar{D}$				

Write your simplified expression here.

- B. Assume you need a **3x8 Decoder** to be used in a circuit. However, you don't have a **3x8 Decoder** in your circuit elements collection. Instead, you have two (02) **2x4 Decoders** with **enable/disable** input and an extra gate from **AND, OR, NAND, and NOR** gate types. Use an appropriate circuit diagram and show how are you going to construct the required **3x8 Decoder** behavior **only** using these available circuit components.

[05 Marks]

- C. Consider a hypothetical sequential logic circuit which consist of four **positive edged D flip flops** as given below.

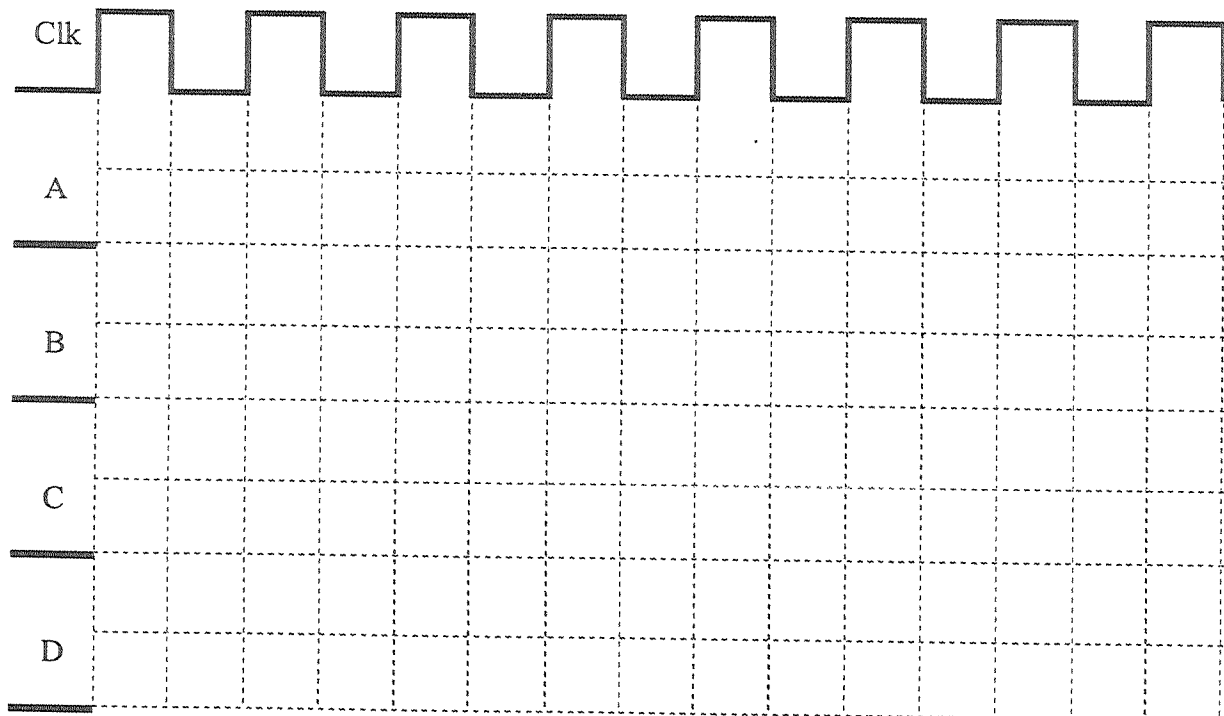


Assume that all A, B, C, and D are in logical state 0 initially (at the time of first clock pulse).

Draw the timing diagram for A, B, C, and D outputs on the dashed line drawing given below based on the available clock pulse pattern.

Note that the initial states of all A, B, C, and D are drawn as state 0 at the first clock pulse.

[05 Marks]



Question 3

A. Briefly describe the relationship between *Clock cycle time* and *Clock frequency* of a CPU.

[02 Marks]

B. What is the *Clock cycle time* for a CPU with 2GHz *Clock frequency*? Show your calculations and steps clearly.

[02 Marks]

C. Briefly describe *Interrupts* and *Exceptions* in a CPU. State a major difference between them.

[03 Marks]

- D. Consider a machine architecture with an instruction set defined as follows. An instruction layout in this machine architecture is described as an instruction is 16 bits long while 4 bits of an instruction is allocated for **OPCODE**. Memory addresses are 8 bits long and a register identifier is also 4 bits long. Following program is executed in this machine. Leftmost column in this set of instructions is the memory addresses that they are stored.

0x40.	LI	R1	0x0A
0x42.	ST	R1	0x80
0x44.	LI	R2	0x00
0x46.	ST	R2	0x82
0x48.	LI	R0	0x08
0x4A.	ADD	R3	R1 R2
0x4C.	LR	R2	R3
0x4E.	LI	R4	0xFF
0x50.	ADD	R5	R1 R4
0x52.	LR	R1	R5
0x54.	JMP	R1	0x58
0x56.	JMP	R0	0x4A
0x58.	ST	R1	0x80
0x5A.	ST	R2	0x82
0x5C.	HLT		

Following are descriptions for some of the *opcodes* in this machine.

LD R A	Load the register R with the content of memory cell with the address A
LI R I	Load the register R with the value I in hexadecimal
LR R1 R2	Load the register R1 with the content of the register R2
ST R A	Store the content of register R to the memory cell whose address is A
ADD R0 R1 R2	Add the two integers in registers R1 and R2 and place the result in R0. Negative numbers are in two's complement form.
JMP R A	Jump to the instruction located in the memory cell with the address A if the bit pattern in R is equal to R0
HLT	Halt the execution

- i. Assuming negative values are in *two's complement* representation, what is the equivalent **Decimal value** that has been loaded to the register **R4** by the following instruction in the program? Justify your answer.

• **LI R4 0xFF**

[01 Mark]

- ii. Based on the defined instruction layout, what is the **maximum** number of registers this machine can have? Give reasons for your answer.

[02 Marks]

- iii. The following instruction in the program may introduce an iterative execution of a subset of instructions within the program.

• **JMP R0 0x4A**

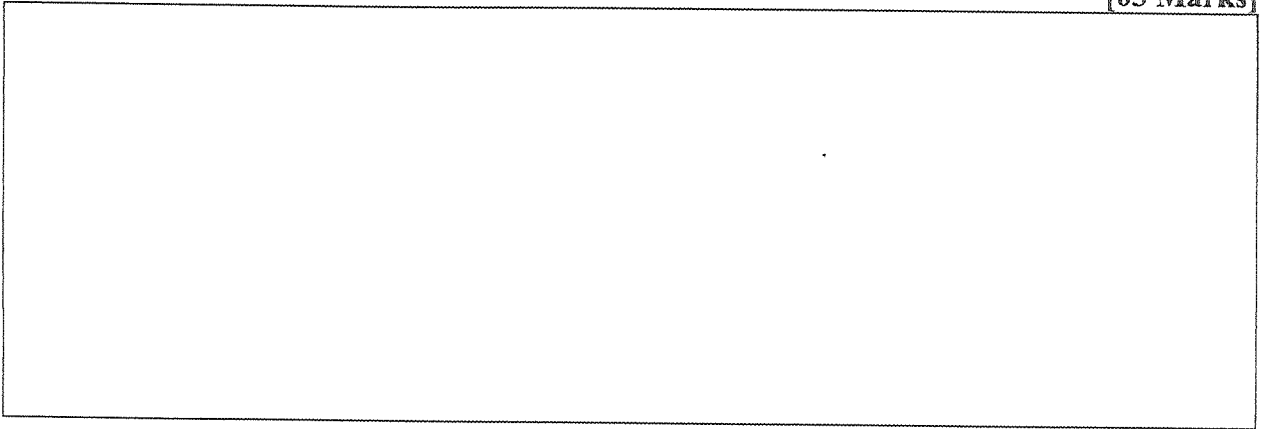
How many times will the subset of instructions be executed due to this **JMP** instruction? Give reasons.

[02 Marks]

Index No:

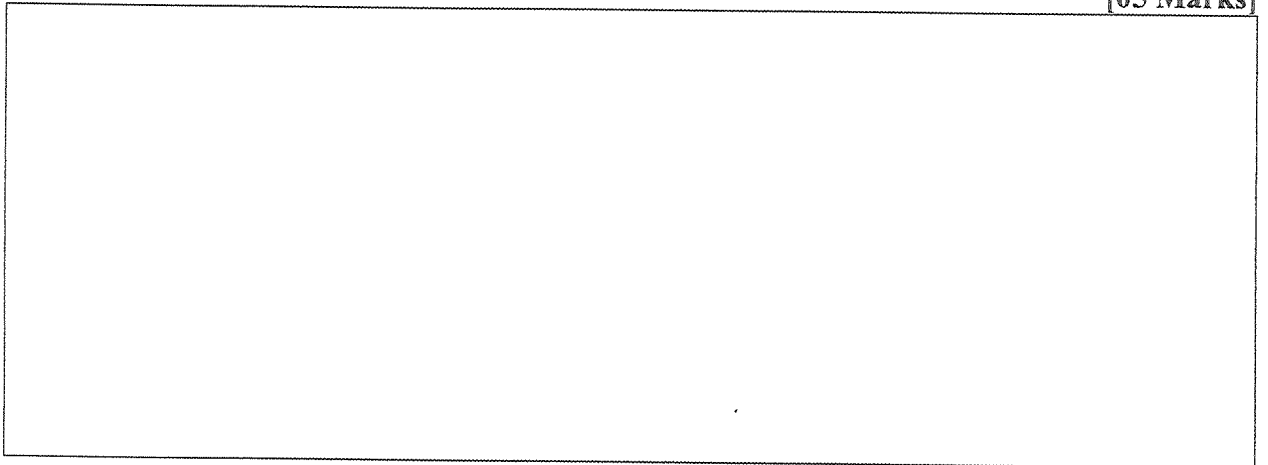
- iv. Briefly explain **two (02)** instructions at the beginning (first two instructions) of the program above and describe what will happen with those two instructions.

[03 Marks]



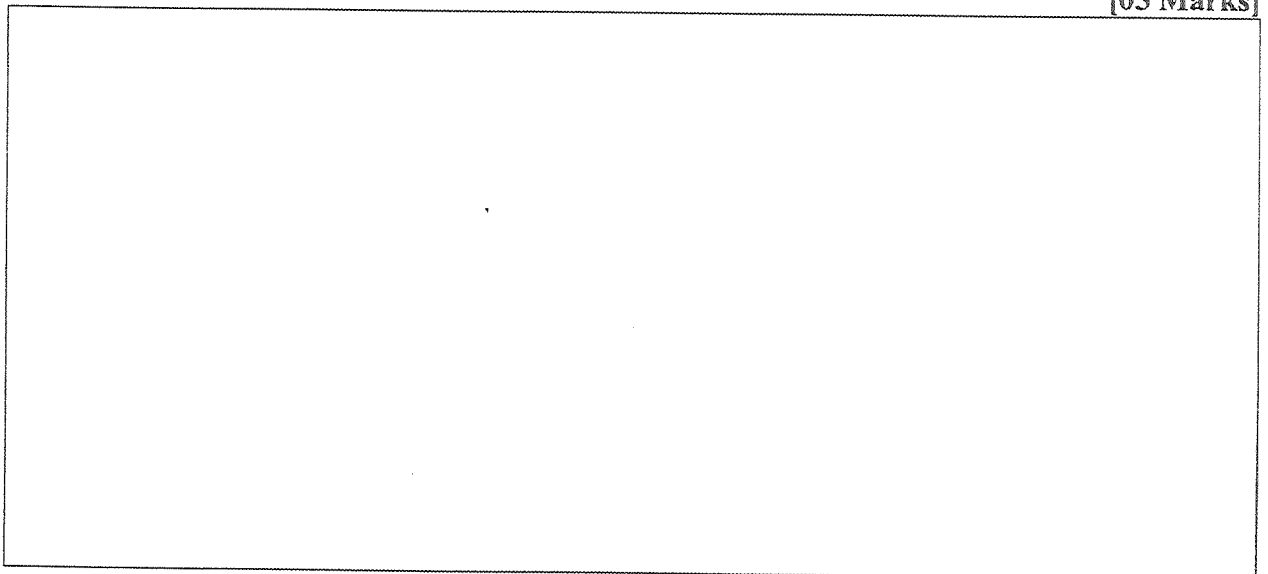
- v. What will be the value stored in memory address **0x80** at the time of CPU executing the final instruction **HLT** of this program? Briefly explain your reasoning behind the answer.

[05 Marks]



- vi. What will be the value stored in memory address **0x82** at the time of CPU executing the final instruction **HLT** of this program? Briefly explain your reasoning for the answer.

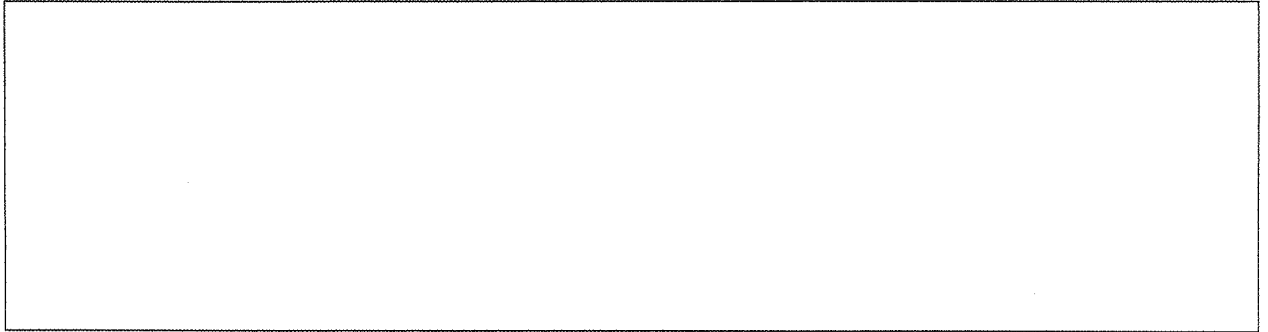
[05 Marks]



Question 4

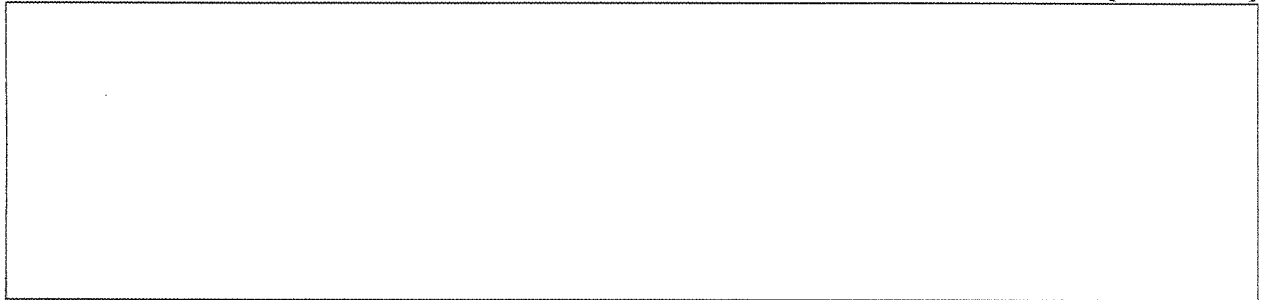
- A. Briefly describe how does a typical cache memory system in a computer confirms *Temporal Locality*?

[02 Marks]



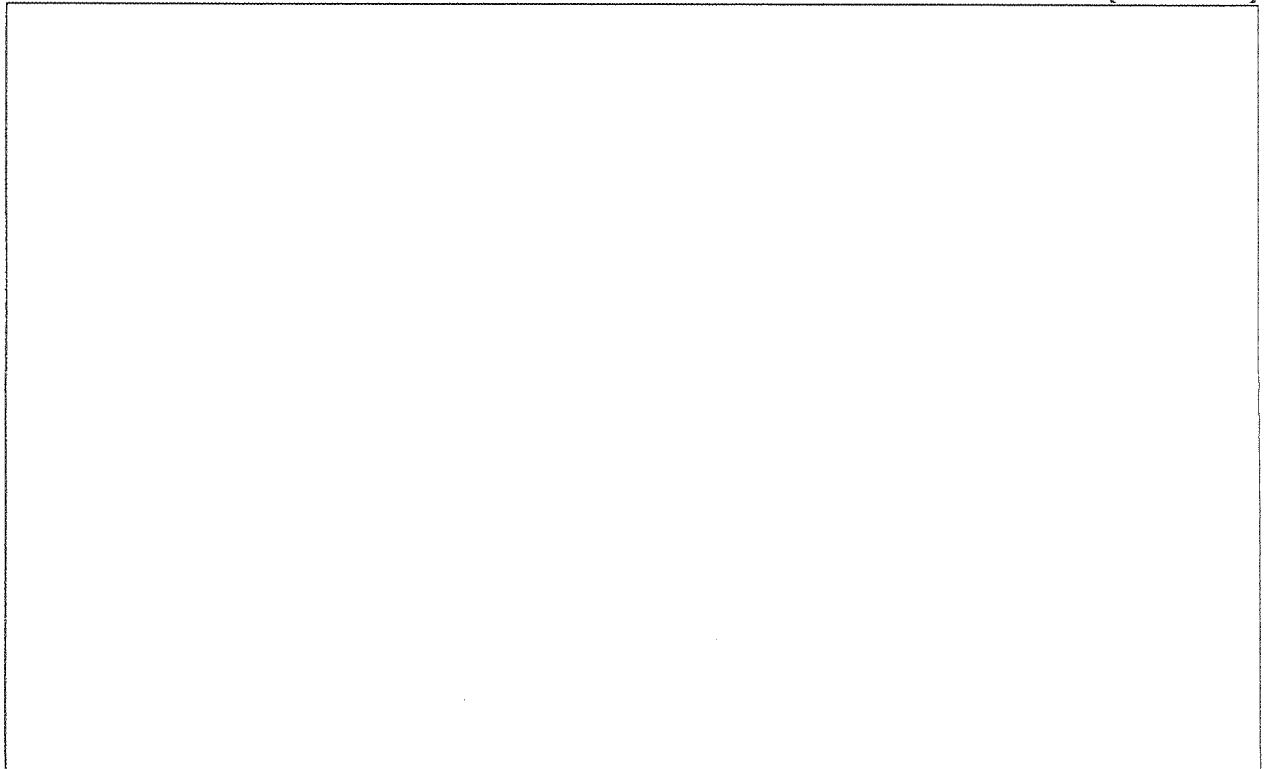
- B. Briefly describe how does a typical cache memory system in a computer confirms *Spatial Locality*.

[02 Marks]



- C. Compare and contrast *Direct-Mapped Cache* and *Fully Associative Cache* systems highlighting their advantages and disadvantages.

[04 Marks]



- D. Answer the following questions considering the cache layout diagram given below. Assume the memory system is **Byte-addressable**. The data values given in the diagram are only for the representation purpose and not in any data representation format. Tag, Index, Offset, and Valid Bit numbers are in binary number representation.

Tag	Index	Memory Space for Data								Valid Bit
		000	001	010	011	100	101	110	111	
1000 1000 100	00	AA	AB	AC	AD	AE	AF	AG	AH	0
2000 2000 200		BI	BJ	BK	BL	BM	BN	BO	BP	0
3000 3000 300		CQ	CR	CS	CT	CU	CV	CW	CX	1
4000 4000 400		D1	D2	D3	D4	D5	D6	D7	D8	1
1000 1000 100	01	EA	EB	EC	ED	EE	EF	EG	EH	1
2000 2000 200		FA	FB	FC	FD	FE	FF	FG	FH	0
3000 3000 300		GA	GB	GC	GD	GE	GF	GG	GH	0
4000 4000 400		HA	HB	HC	HD	HE	HF	HG	HG	1
1000 1000 100	10	IA	IB	IC	ID	IE	IF	IG	IH	1
2000 2000 200		JA	JB	JC	JD	JE	JF	JG	JH	0
3000 3000 300		KA	KB	KC	KD	KE	KF	KG	KH	1
4000 4000 400		LA	LB	LC	LD	LE	LF	LG	LH	0
1000 1000 100	11	MA	MB	MC	MD	ME	MF	MG	MH	1
2000 2000 200		NA	NB	NC	ND	NE	NF	NG	NH	1
3000 3000 300		OA	OB	OC	OD	OE	OF	OG	OH	0
4000 4000 400		PA	PB	PC	PD	PE	PF	PG	PH	1

- i. What is the size of a *cache line* in this cache memory system? Show reasons and calculations clearly.

[02 Marks]

- ii. What is the size of the entire cache memory according to the diagram? Show your reasons and calculations clearly.

[02 Marks]

- iii. If the cache memory given in the above diagram in Question 4.D is assumed as a *K-Way Set Associative* cache, identify the most appropriate number for *K* that corresponds to the cache layout in the diagram. Justify your answers with reasoning.

Note that a *Direct Mapped* cache can be seen as a *K-Way Set Associative* cache when $K = 1$.

[03 Marks]

- iv. Assume that the cache system applies *Least Recently Used* (LRU) cache replacement policy where it discards the least recently accessed cache entry when it needs to accommodate a new cache line. Then the CPU refers following three (03) addresses in a sequence when its cache memory is as given in the above diagram.

- 1000 1000 1001 0010
- 2000 2000 2001 0010
- 2000 2000 2001 0011

State whether each address reference would result a cache miss or cache hit by identifying **Tag**, **Index** and **Offset** components of an address. Justify your answer with reasons for each case.

[06 Marks]

1000 1000 1001 0010

2000 2000 2001 0010

2000 2000 2001 0011

- v. Immediately after referring the above three (03) addresses in question 4.D.iv, the CPU refers following three (03) addresses in a sequence in the same cache memory system.
Note: The cache memory employs *Least Recently Used* cache replacement policy.

- 5000 5000 5001 0011
- 4000 4000 4001 0011
- 3000 3000 3001 0011

State whether last two (02) addresses in the list would result a *cache miss* or *cache hit*. Justify your answers with reasons.

[04 Marks]

4000 4000 4001 0011

3000 3000 3001 0011
